STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

TEAM ID:	TEAM NAME:
LRC-26-0143	NIHAO

Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable**. An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT

Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN

Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	**8mm**
Fluid density (ρ) =	**1000kg/m^3**
Dynamic Viscosity (μ) =	**1x10^{-3}Pa.s**
Required Flow range =	**1-6 L/min**
Nominal target flow for the live challenge =	**3L/min**

Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity 2. Pumps – 3 Quantity 3. FT-01 Flow Transmitter – 01 Quantity 4. Ball valve – 3 Quantity 5. ON/OFF/XV Valve – 1 Quantity 6. 10-mm OD tubing/blue – 9 meters/1 meter 7. Microcontroller/PCB/ data-acquisition software 8. Required electrical/ electronic accessories	1. Custom PCB 2. ESP32 NodeMCU module 3. Pump Driver-1 4. Pump Driver-2 5. Pump Driver-3 6. 12 V DC power input (**SMPS**) 7. 12 VDC Solenoid-valve interface 8. Flow-sensor interface 9. Float-switch interface 10. ON/OFF-switch interface 11. Pump-driver PWM/enable interfaces 12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code**. The case-1 can be turned on using the available on/off Switch, to start the system.

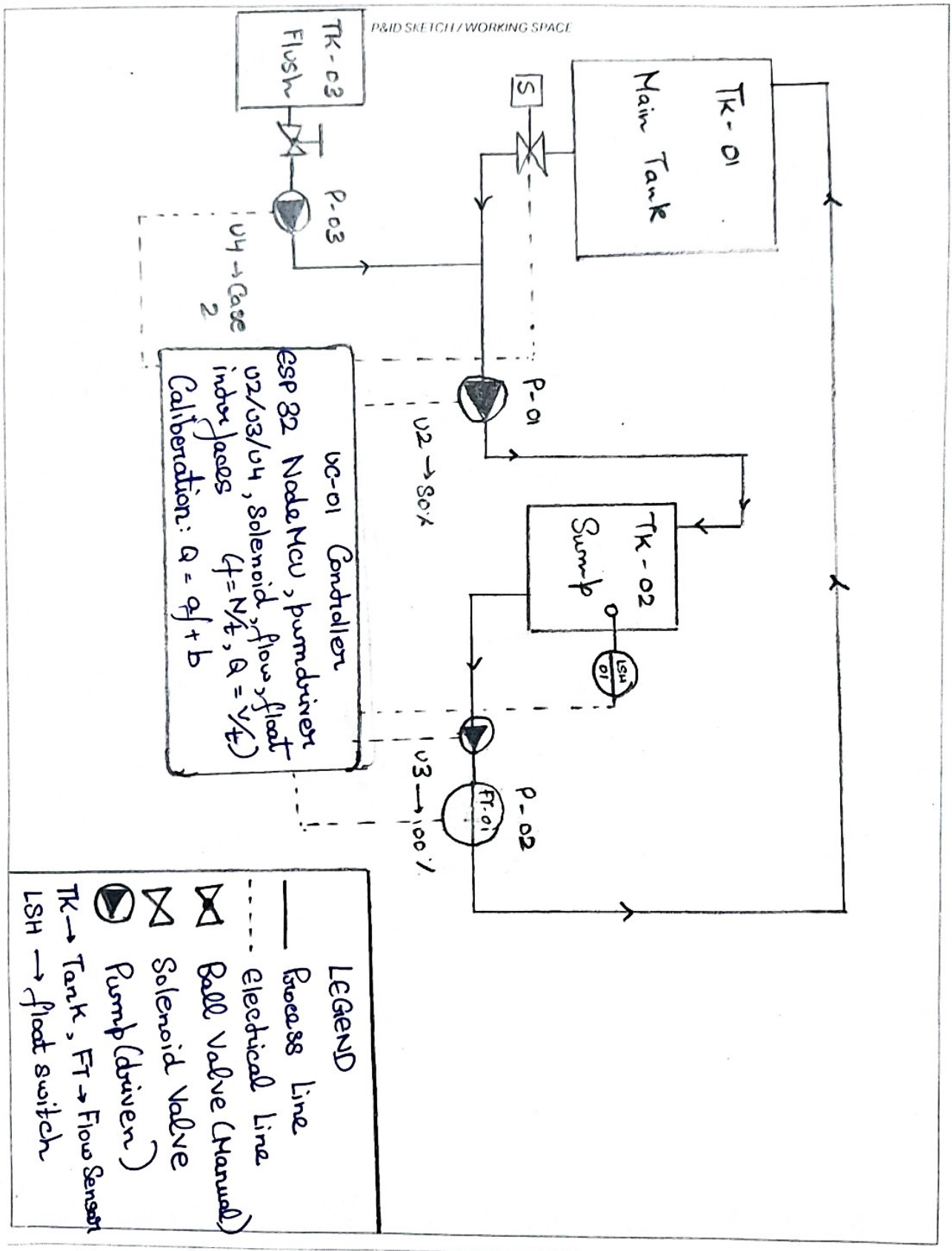
CASE 2 – Flush/Line Conditioning Sequence

Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

SUBMIT:

A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

B. Final P&ID Block diagram


P&ID SKETCH / WORKING SPACE
TK-03 Flush
P-03
V4 → Case 2
S
TK-01 Main Tank
P-01
V2 → 50%
TK-02 Sump
LSH 01
V3 → 100%
FT-01
P-02
UC-01 Controller
ESP 32 NodeMCU, pumdriver
V2/V3/V4, Solenoid, flow, float
interfaces
Calibration: Q = qf + b
LEGEND
Process Line
Electrical Line
Ball Valve (Manual)
Solenoid Valve
Pump (driven)
TK → Tank, FT → Flow Sensor
LSH → float switch

Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

WORKING / COMPARISON

Theoretical velocity $\Rightarrow V = Q/A$, $A = \pi (0.008)^2/4 = 5.03 \times 10^{-5}\ m^2$

Using the nominal flow, Actual velocity used same formula with measured flow

$R_e = \rho V D/\mu = 8000 \cdot V$

1 L/min → Theoretical (m/s) → 0.332

R_e → 2650

Measured flow = 1.00 ⇒ Actual velocity → 0.332

Deviation → 0%.

3 L/min → Theoretical (m/s) → 0.995

R_e → 7960

Measured flow = 2.8 ⇒ Actual velocity → 0.928

Deviation → − 6.7%.

6L/min → Theoretical (m/s) → 1.989

R_e → 15900

Here the pump couldn't Reach more than 5L/min thus we don't have the right data.

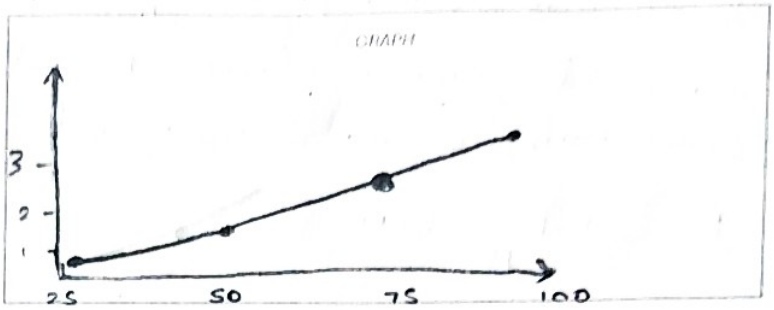
for the first the behaviour is transitional where as for the other 2 it's turbulent.

Actual velocity falls below due to friction & minor losses.

FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.38
50%	1.13
70%	2.29
100%	2.91



Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

Report:

R^2 =	0.989
a =	0.01088 L/min per Hz
b =	−0.823 L/min
Maximum calibration error =	0.086 L/min (About 8%)
Average calibration error =	0.059 L/min (About 3.6%)

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	3867	30	128.9	0.5	1
2	5454	30	181.8	0.87	1.74
3	6900	30	230.0	1.12	2.24
4	7539	30	251.3	1.2	2.4
5	8775	30	292.5	1.40	2.80

CALCULATIONS / OBSERVATIONS

Formulas used

$f = N/t$, $Q = V/t \times 60$ (t in s)

Caliberation fit: $Q = a \cdot f + b$

Caliberation Error: Q_fit - Q_actual in L/min & as % of Q_actual.

Sample calculations

$N = 3867$ pulses, $t = 30$ s, $f = 3867/30 = 128.9$ Hz

$V = 0.5$ L, $Q = 0.5/30 \times 60 = 1.00$ L/min

fit $Q = 0.01088f - 0.323$

$R^2 = 0.989$ ⟶ max Error 7.9 %.

observations:-

1. Flow rises linearly with pulse frequency.
2. The sensor over-reads the Collected Volume especially at Low flow.
3. Pulses per Litre (K-factor) is about 6K ~ 7K
4. Error rising in Low flow & Reduces in high.
5. Tested Range 1 ⟶ 2.8 L/min

HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

ANSWER Here the formula is $T_r = V/Q$, not Q/V ~ this gives freq.

$V = 0.5L$, $Q = 2.5L/min$, $T_r = V/Q = 0.5/2.5 = 0.2 min = 12s$

1. It's the avg time a fluid spends inside the tube from beginning to ending, also the time a change at the inlet (a pump change, valve movement / blockage) takes to travel through the section & show up at the outlet/sensor

2. If $V \rightarrow 2V \Rightarrow T_r \rightarrow 2T_r$ Since $V \propto T_r$. The System Responds more sloudy, more Lag.

3. If $Q \rightarrow 2Q \Rightarrow T_r \rightarrow T_r/2$ Since $T_r \propto 1/Q$. Fluid moves faster & changes shown up sooner.

4. • The system has a hydraulic lag, change in pump doesn't appear at FT-01 instantly. Appears after roughly one residance time.
   • Reading the FT too early can lead to a wrong diagnosis.
   • Set's how Long to wait before recording a stable reading, or detection of fault avoiding false alarms
   • Matters in flushing time too since it takes several residence times to fully displace the old process fluid.

Q3.2 The following observation are obtained from the same system:

CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

ANSWER Condition C & almost Condition E too

In C FT-01 Reads 0. It cannot say why because the same Reading is produced by a blocked Line, a failed pump/sensor. FT-01 only reports pulses & "no pulse" is ambiguous. A truly dead sensor / truly dead flow look identical from Sensor's pov.

E is Similar, Intermittent pulses could mean real flow fluctuation or just noisy/loose Signal.

Addition Info needed to tell them apart.

- Pump status feedback → pump motor current view, RVM/ enable state or listening for pump sound/vibration. This shows weather the pump is actually running or has failed.
- Pressure measurement: pressure sensors upstream/downstream of the valve/flow transmitter.
- Level data from the tank/sump → Confirms whether there is fluid to pump & whether Sump lvl is changing which proves flow independently of FT-01

others are Sensor diagnostics, visual & physical inspection and trend history & response to Commands.

HARDWARE LIST

Mechanical Hardware

Component	Description	Main Use
Ball Valve (BV-01)	A manually operated valve with a spherical shut-off mechanism.	Used to **manually isolate or permit fluid flow** in the pipeline. It is useful for maintenance, testing, and flow isolation.
10-Litre Fluid Tank (Fluid Tank-01)	Transparent storage tank with approximately 10 L capacity.	Used as the **main fluid reservoir** for storing process fluid. The transparent body also allows the fluid level and condition to be visually monitored.
5-Litre Sump	Small transparent reservoir labelled **SUMP**, approximately 4.5 L.	Used as a **collection/storage vessel** for fluid returning to or entering the sump section of the system.
4.2-Litre Flush Fluid Tank / Flush Line	The transparent vessel labelled **FLUSH FLUID** is approximately 4.5 L.	Provides **flush fluid** for cleaning or flushing the main pipe circuit when required.
ON/OFF Valve (OV-01)	Electrically operated valve/solenoid-type valve shown in the setup.	Used for **automatic control of fluid flow**, allowing the controller to turn a particular flow path on or off.
Pumps -Pump-1, Pump-2, Pump-3	Motor-driven fluid pumps. Three pumps are shown in the setup.	Used to **circulate and transfer fluid** between different sections of the system. Each pump can be independently controlled.
8 mm Process Pipe	Flexible blue tubing with approximately 8 mm process-line size.	Used as the **main fluid-flow path** connecting tanks, valves, pumps, tees, and other components.
10 mm × 8 mm Reducers	Reducer fittings that transition between a 10 mm connection and an 8 mm pipe.	Used to **adapt the pump connection size to the 8 mm process pipe. Mainly used for Pumps**
8 mm Connectors	Push-fit/barbed-type fittings compatible with the 8 mm tubing.	Used to **join pipes to components** and to make detachable connections between sections of tubing.
8 mm Equal Tee	T-shaped connector having three equal 8 mm ports.	Used to **split one 8 mm fluid line into two branches or combine two branches into one line.**

Electronic Hardware

Electronic Component	Identification / Function	Use
PCB -Printed Circuit Board	Custom control board	Provides the electrical interconnection between the **12 V power supply, ESP32 controller, sensors, valves, indicators, and pump-driver interfaces.**
ESP32 / NodeMCU ESP32S module (U7)	Main microcontroller	Acts as the **central controller.** It provides GPIO/PWM signals for controlling outputs and receiving signals from sensors/switches.

Electronic Component	Identification / Function	Use
12 V DC Input Connector	Marked **12V INPUT**	Connection point for the external 12 V DC power supply.
12 V Output Connectors	P1, P2, P3 and additional 12 V output terminals	Provide 12 V power connections for external loads such as the pump-driver/motor circuits.
Motor Driver Interfaces -U2, U3, U4	Three multi-pin driver connectors shown on the PCB	Provide dedicated interfaces for **Pump-1, Pump-2 and Pump-3 driver/control connections**.
Solenoid Valve Connector	Labelled **SOLENOID_VALVE -S/V/G**	Interface for controlling/connecting the solenoid or ON/OFF valve.
Flow Sensor Connector	Labelled **FLOW_SENSOR - S/V/G**	Interface for the water-flow sensor; supplies power and receives the sensor signal. **THE FLOW SENSOR SHOULD BE CALIBRATED**
Float Switch Connector	Labelled **FLOAT_SWITCH -FSW/GND**	Interface for a float/level switch used to detect a water-level condition.
ON/OFF Switch Connector	Labelled **ON/OFF_SWITCH -SW/GND**	Interface for an external manual ON/OFF switch.
12V DC SMPS	**12V DC SMPS**	For Power Supply to the board